CSA15 Cloud Computing and Big Data Analytics

Capstone Cloud Technical Presentation

Individual Student Presentation Deck

Presenter Details

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Class / Section / Slot:C

Capstone Title:AquaGuard Cloud Water Quality Reporter With Sensor Analytics and Public Alerts

Selected SDG Goal(s):SDG 6 – Clean Water and Sanitation

Cloud Platform / Tools:Firebase, Google Cloud, ThingSpeak (or AWS IoT)

Problem Context and Stakeholder Mapping

Our capstone begins with a real problem to solve

Problem Statement

Many communities lack an easy way to monitor and report drinking water quality. AquaGuard provides a cloud-based platform for water quality monitoring and public safety alerts.

Target Users

- Citizens
- Water authorities
- Health departments
- NGOs

SDG Fit

- **SDG 3:** Good Health and Well-being
- **SDG 6:** Clean Water and Sanitation
- **SDG 11:** Sustainable Cities and Communities

Product Boundary

- Android app & web dashboard
- Cloud database & analytics
- Public alert system

Cloud Adoption Rationale and Concept Map

The problem needs cloud support to become a usable product

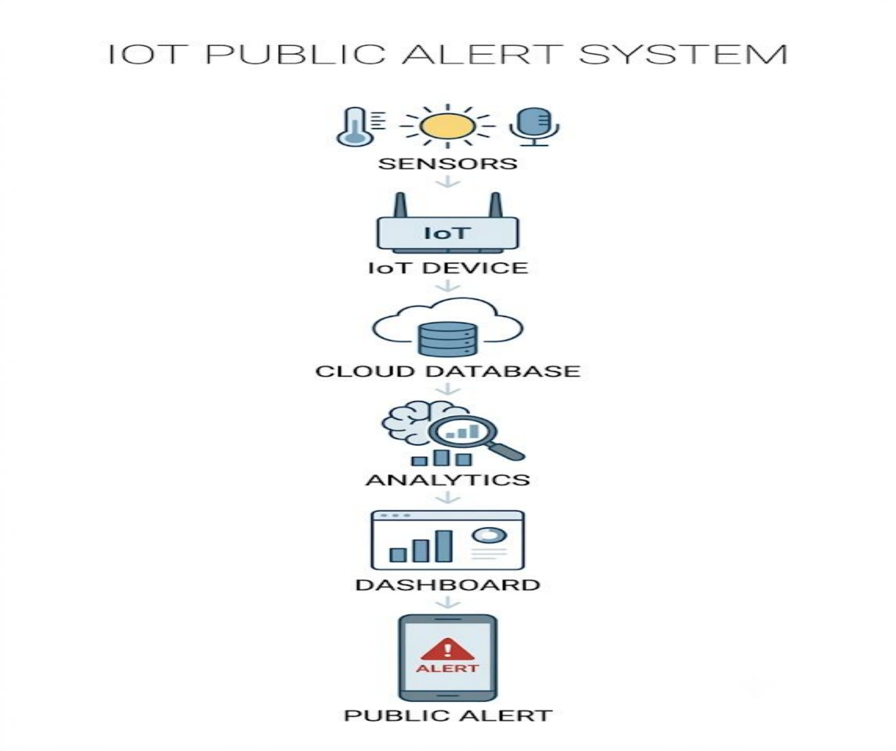
Cloud Definition

Cloud computing provides online storage, processing, and analytics for real-time water quality monitoring.

Cloud Characteristics

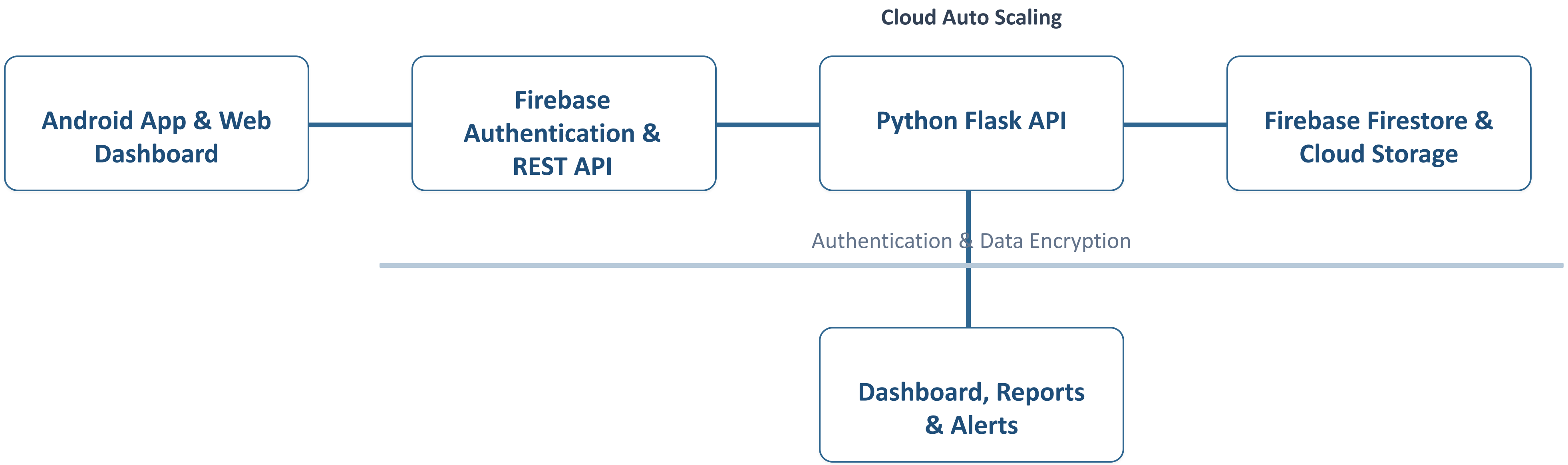
- Scalability
- Elasticity
- High Availability
- Measured Service

Cloud Concept Map



Capstone Cloud Architecture

Our cloud architecture connects the user action to services



Cloud Service Model Responsibility Matrix

Each service model defines what we control and what cloud manages

Service area	Model	Provider manages	Student team manages	Why this fits
Frontend hosting	SaaS	Hosting	UI Design	Easy access
Backend/API	PaaS	Runtime	APIs	Fast deployment
Database	DBaaS	Backup	Data	Secure storage
Storage	Cloud Storage	Files	Uploads	Reliable storage
Monitoring / logs	SaaS	Logs	Dashboard	System tracking
Security / IAM	IAM	Authentication	User Roles	Secure access

Data Flow and Analytics Pipeline

The product data moves from user activity to useful insight

User reports / Sensor data — REST API — Firebase Database — Python Analytics — Dashboard & Charts

Data Requirements

Water quality readings, user reports, location

Analytics Requirement

Water quality score, trends, public alerts

Proof Required

Sample report and dashboard result

Python / AI Module Integration

Python or AI adds intelligence where the product needs it

Python Role

Python analyzes water quality data and automatically generates reports and alerts.

Input and Output

The system receives pH, TDS, turbidity, and sensor values, then displays the water quality status.

Execution Point

The Python module runs through a cloud API whenever new water quality data is submitted.

Product Value

It improves decision-making by providing fast, accurate, and automated water quality analysis.

Security, Scalability, Privacy and Cost Design

Security, scale, privacy and cost shape the final design

Security

User authentication, encrypted data transfer, and secure APIs protect the system.

Privacy

Only authorized users can access water quality reports and sensitive information.

Scalability

The cloud platform can support more users and future IoT sensor integration without redesign.

Cost Awareness

The project uses free-tier cloud services and open-source tools to reduce deployment costs.

Implementation Stack and Deployment Evidence

Our implementation stack shows how the design will be built

Mobile / Web Layer

The application is developed using Flutter for Android and a responsive web dashboard.

Cloud Services

Firebase provides authentication, cloud storage, database services, and real-time synchronization.

Evidence

The project is demonstrated using the Firebase Console, dashboard screenshots, and GitHub repository.

Version Control

GitHub is used to manage the source code, track changes, and support team collaboration.

Demo Storyboard and User Journey

A complete user journey proves the product flow

Scene 1

The user logs into the application using a secure authentication system.

Scene 2

The user uploads water quality data or sensor readings through the mobile app.

Scene 3

The cloud stores the data and the Python module analyzes the water quality.

Scene 4

The dashboard displays reports, trends, and water quality status in real time.

Scene 5

If unsafe water is detected, the system sends an alert notification to users.

Scene 6

Authorities monitor reports through the web dashboard and take necessary action.

Evaluation Readiness Checklist

Our evidence confirms that the presentation is ready

- ✅ Cloud ecosystem and service model explained
- ✅ Elasticity and scalability decisions are visible
- ✅ REST/SOAP/API role connected to capstone
- ☐ GitHub and cloud evidence are inspectable
- ✅ Architecture has compute, data, storage, security, monitoring
- ✅ References and integrity statement are included

Conclusion, Limitations and Future Scope

We close with learning, limitations and next steps

Learning Outcome

This project improved our understanding of cloud computing, APIs, databases, and real-time analytics.

Limitations

- Internet connection is required.
- IoT sensor integration is optional.
- Limited to basic water quality parameters.

Future Improvement

- AI-based water quality prediction.
- GPS-based water source mapping.
- Real-time IoT sensor integration.
- Multi-language support.

Integrity Statement

This project was developed for academic purposes using Flutter, Python, Firebase, GitHub, and publicly available learning resources.